Least privilege and tool scopes

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Make each granted power specific and each denied path observable.

Use exact pairs, actor/token intersections and ledger checks; keep current catalogue assessment distinct from older practice.

Keep one habit: Model a denied call → inspect allowed/denied traces → diagnose overgranting → assess fresh catalogue reasoning → revisit a narrow trusted expansion.

MAIN LESSON · SECTION 1

Prepare the connected service investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of **least privilege** and tool scopes

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Authentication and authorisation. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of authentication and authorisation

Reference example, not the only acceptable wording: Teacher explanation: Authentication identifies an actor; authorisation decides whether that actor may perform a particular operation on a particular resource.

Use trusted session roles, never a role claimed inside user JSON. Student may read student resources; teacher may read both. Unauthenticated requests are denied.

Without opening the old lesson, explain one key idea from Typed contracts and schemas. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of typed contracts and schemas

Reference example, not the only acceptable wording: Teacher explanation: Typed contracts define values, required fields and failure outcomes. Runtime **input** still needs validation even when program annotations exist.

Accept exactly room_id as a nonempty string and requested_count as a nonnegative integer excluding bool. Reject extra fields, missing fields and wrong types before **process**ing.

Pictured end goal

Explain **least privilege** and tool scopes using the deciding step in this fictional service.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Teach the learning cycle

Case exposure check: this source example appears in ai-2/31, ai-6/6. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.
</details>

Retrieval answer guidance

Authentication and authorisation

Typed contracts and schemas

Use the three-session teaching cycle above, with extra time as needed. **Least privilege** grants only capabilities required by the task. Tool access should include resource and operation scope, not merely a broad tool name. Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of **least privilege** and tool scopes

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach exact scope pairs and stopping before execution

> Keep the selected catalogue case separate from older inventory practice.

Definitions and complete actor/capability contract

Least privilege grants only capabilities required by the task. Tool access should include resource and operation scope, not merely a broad tool name.

A reporting assistant may lookup public rooms but may not write records, send messages or read teacher notes. Check actor, operation and resource before executing a proposed call.

Main reporting-assistant contract	Supplied scope
Trusted actor	reporter
Actor policy	Only the pair (lookup, public-room)
Current token capability	Only the pair (lookup, public-room)
Proposed excluded call	(write, teacher-notes): wrong operation and wrong resource
Permitted comparison call	(lookup, public-room), subject to other required checks
Execution boundary	Denied proposal must not reach the execution stub
Scope of evidence	Local immutable permission sets and a recorded stub; no real token or tool execution

Main mismatches, event trace and checked reasoning

Check the exact pair before a proposal can execute

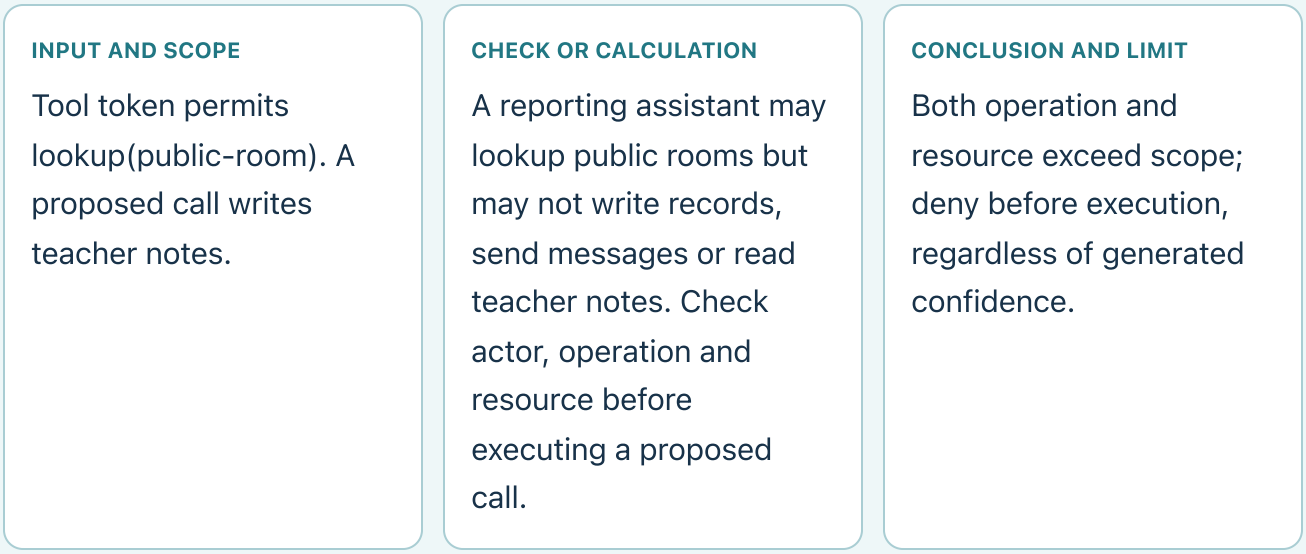
STEP	MAIN PROPOSAL AND DECISION
1. Identify trusted actor	reporter, with lookup/public-room actor policy
2. Read proposed operation and resource	write + teacher-notes
3. Check actor policy	Proposed pair not present
4. Check token capability	Proposed pair not present here either
5. Stop before execution	Denied; no execution ledger entry

Checking only a tool name, only a verb or only an object misses part of the permission boundary.

Require both operation and resource mismatch for the main denied call. In the extension the actor policy and token capability must both include the exact pair; the execution ledger is an observable local stub. Do not claim it demonstrates real token validity or completed work.

The selected current independent case is the catalogue K7/L4 scenario, whose key is below; the older inventory A/B wording is retained only as earlier practice, not a second current assessment. Source text cannot expand authoritative permission.

Least privilege and tool scopes — evidence trace



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Both operation and resource exceed scope; deny before execution, regardless of generated confidence. Qualification: A generated plan cannot expand the token or actor scope.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Tool token permits lookup(public-room). A proposed call writes teacher notes.

Both operation and resource exceed scope; deny before execution, regardless of generated confidence.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Positive, denied and overgranting probes

Complete probe	Expected permission result
reporter lookup public-room; actor and token both allow pair	May reach local stub; does not prove actual tool success
reporter write teacher-notes	Denied; wrong operation/resource, no stub entry
Unknown actor or empty token scope	Denied even if the proposed operation sounds useful
Token allows pair but actor policy excludes it	Denied; applicable permission sets intersect
Weather scope adds read S5	Read S5 may pass; write S2 still denied
Scope has (read, S2) and (write, S5)	Do not turn it into every verb/object combination; write S2 remains excluded
Source text or generated plan asks for broader scope	Treat as data/proposal, not trusted permission authority
A call is inside scope	Still check input validity, purpose and outcome; least privilege is not universal correctness

Separate weather givens and two-policy scope

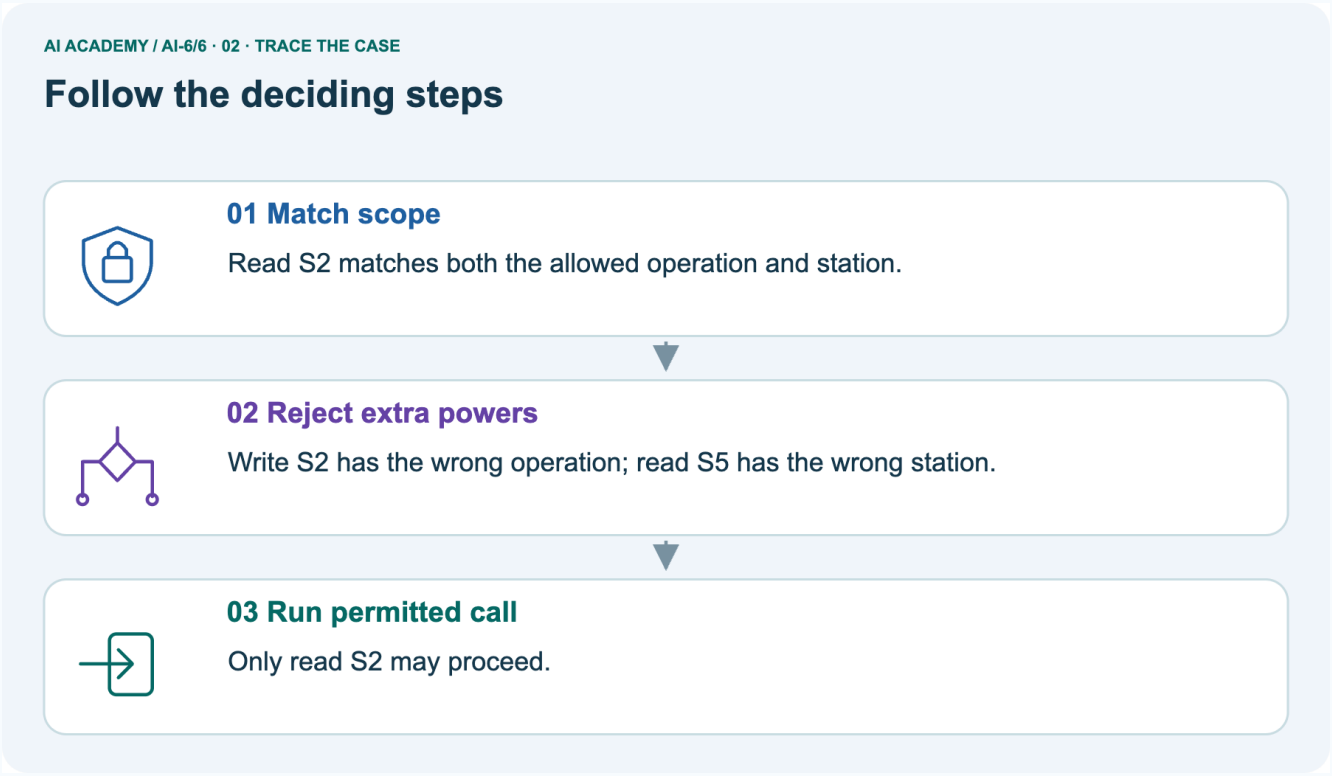
A narrowly scoped weather tool. A school bot holds a token allowing only reading station S2.

Requested calls: read S2, write S2, read S5. Trusted scope is read on S2 only.

The main room-reporting example uses lookup/public-room. The picture is a separate weather service with read/S2 permitted. Read S2 matches both operation and station; write S2 fails the operation; read S5 fails the station. If trusted scope explicitly adds read S5, that pair can pass while write S2 remains excluded.

The extended local program also declares an actor policy allowing read S2/S5, then narrows it with a token allowing S2 only. Expanding a token cannot overcome an actor policy that still forbids the pair. No weather reading or real token validation occurs.

Weather trace and added-read-station key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Match scope	Read S2 matches both the allowed operation and station.
2. Reject extra powers	Write S2 has the wrong operation; read S5 has the wrong station.
3. Run permitted call	Only read S2 may proceed.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Least privilege limits a component to needed powers. Checking only the verb or only the object would leave a gap.

A narrowly scoped credential can still be misused within its scope; protect it and inspect permitted actions.

Change: Add read S5 explicitly to the trusted scope.

Prediction: Read S5 becomes permitted; write S2 remains denied.

Reason: Adding one station does not grant a write operation.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Least privilege	Granting only the necessary authority
Capability	A scoped permission to perform an operation
Resource scope	The specific records an operation may address

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

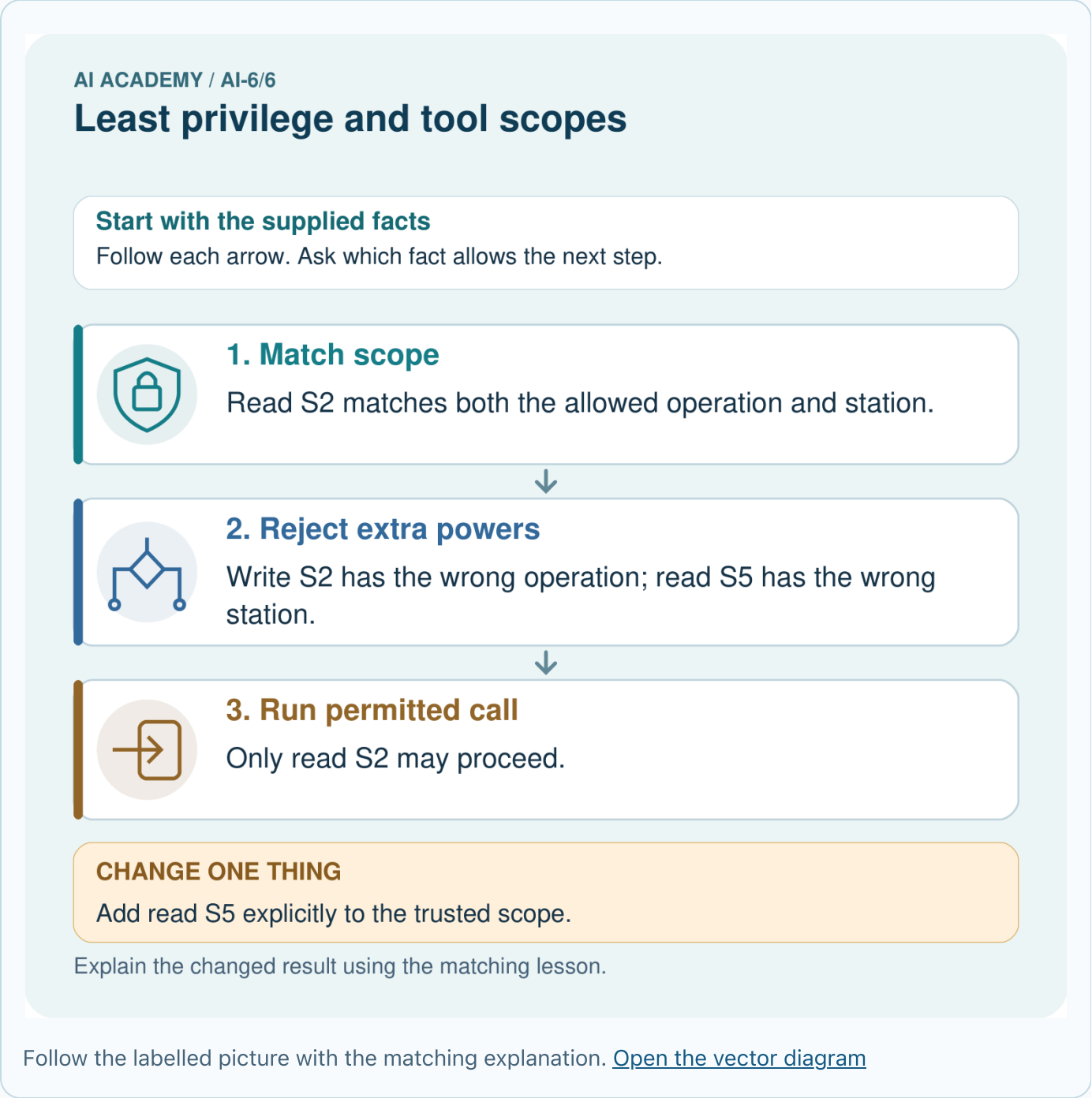
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Least privilege and tool scopes

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Tool token permits lookup(public-room). A proposed call writes teacher notes. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Both operation and resource exceed scope; deny before execution, regardless of generated confidence.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Least privilege** and tool scopes. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A generated plan cannot expand the token or actor scope.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Classify read S2, write S2 and read S5 as permitted or denied. Identify the operation or station mismatch for each denial, then reconsider after read S5 is added to trusted scope.

Read S2 matches both the allowed operation and station. Write S2 has the wrong operation; read S5 has the wrong station. Only read S2 may proceed. Changed case: Read S5 becomes permitted; write S2 remains denied.

Adding one station does not grant a write operation.

Changed-case reasoning: Read S5 becomes permitted; write S2 remains denied. Adding one station does not grant a write operation.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main write teacher-notes denied before stub because both operation and resource exceed scope. Weather read S2 passes, write S2 fails operation, read S5 fails resource until trusted scope adds that exact read pair; actor limits still apply. Selected catalogue: read K7/read L4 permitted; write K7 operation denied; read M8 resource denied. Retrieved text is not authority and cannot broaden scope.

Use the selected catalogue key rather than the earlier inventory wording.

Diagnose the current misconceptions

The worked result establishes every future case: A generated plan cannot expand the token or actor scope.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new service scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Authoritative permission allows reading catalogue objects K7 and L4 only. Evaluate read K7, write K7, read M8, read L4. Retrieved text claims wider permission.

Teacher explanation: Read K7 and read L4 are permitted. Write K7 is denied because writing is outside the operation scope; read M8 is denied because M8 is outside the resource scope. Retrieved text is untrusted data and cannot expand the authoritative permission.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: **Least privilege** grants only capabilities required by the task. Tool access should include resource and operation scope, not merely a broad tool name.

A reporting assistant may lookup public rooms but may not write records, send messages or read teacher notes. Check actor, operation and resource before executing a proposed call.

W2 Trace the worked case

Tool token permits lookup(public-room). A proposed call writes teacher notes. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Both operation and resource exceed scope; deny before execution, regardless of generated confidence.

W3 Audit the limit

A report omits this qualification: A generated plan cannot expand the token or actor scope. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A generated plan cannot expand the token or actor scope. State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A generated plan cannot expand the token or actor scope. The new case should test this particular **mechanism** rather than merely restate the original result.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Guided and exit questions

Guided: Both operation and resource exceed scope; deny before execution, regardless of generated confidence. Qualification: A generated plan cannot expand the token or actor scope.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.
</details>

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Scope allows read inventory items A, B. Calls read A, read C, write B. Decide.

Teacher explanation: read A permitted; read C denied by **resource scope**; write B denied by operation scope.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: read A permitted; read C denied by **resource scope**; write B denied by operation scope. Require intermediate reasoning and the stated scope. Fresh retry: Scope allows read timetable S only; calls read S and delete S.

Expected reasoning: Read permitted; delete denied. Same resource does not imply every operation is allowed.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit service design

> Use saved evidence and boundary decisions to choose the next step.

Project and next-week bridge

Add a labelled **least privilege** and tool scopes **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **least privilege** and tool scopes.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Testing pure components. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Delayed return and support record

Suggested return points in this level: ai-6/7, ai-6/9, ai-6/12. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	

Record	Teacher evidence
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of **least privilege** and tool scopes

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Tool token permits lookup(public-room). A proposed call writes teacher notes. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Both operation and resource exceed scope; deny before execution, regardless of generated confidence.

Check every learner before discussion: Use the worked case for **Least privilege** and tool scopes. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A generated plan cannot expand the token or actor scope.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Add read S5 explicitly to the trusted scope. Require a **prediction** and a reason before comparing. Expected result: Read S5 becomes permitted; write S2 remains denied.

Reason: Adding one station does not grant a write operation.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

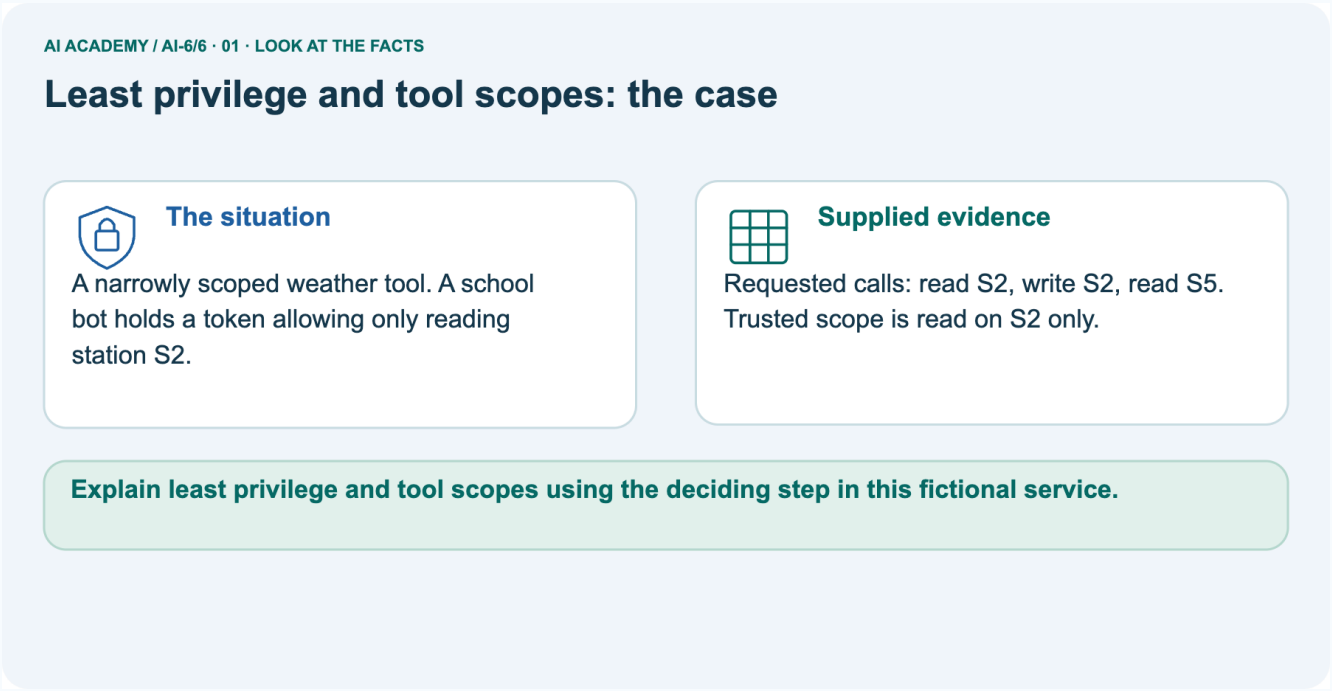
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Least privilege and tool scopes: the case



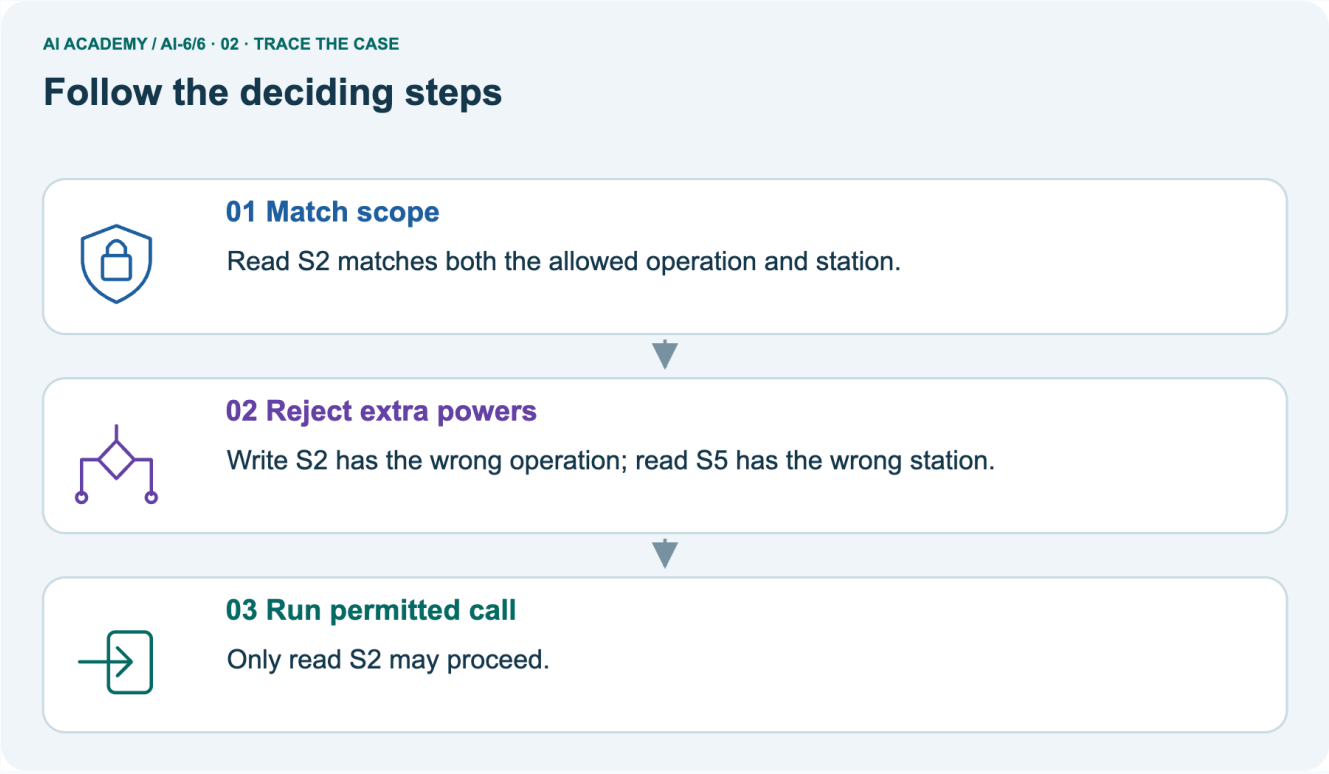
Follow the labelled picture with the matching explanation. Open the vector diagram

A narrowly scoped weather tool. A school bot holds a token allowing only reading station S2. Supplied facts: Requested calls: read S2, write S2, read S5. Trusted scope is read on S2 only.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



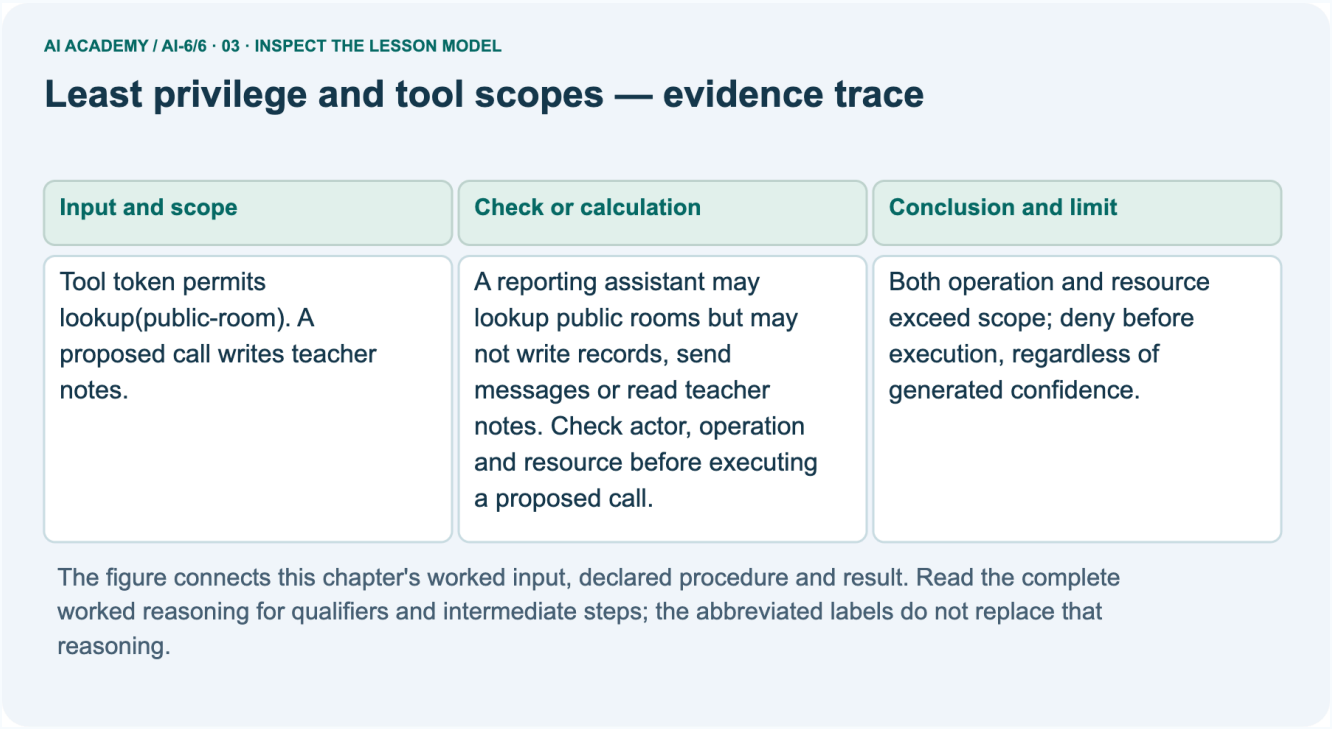
Follow the labelled picture with the matching explanation. Open the vector diagram

Match scope: Read S2 matches both the allowed operation and station. → Reject extra powers: Write S2 has the wrong operation; read S5 has the wrong station. → Run permitted call: Only read S2 may proceed.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Least privilege and tool scopes — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Least privilege and tool scopes — **evidence** trace. Tool token permits lookup(public-room).

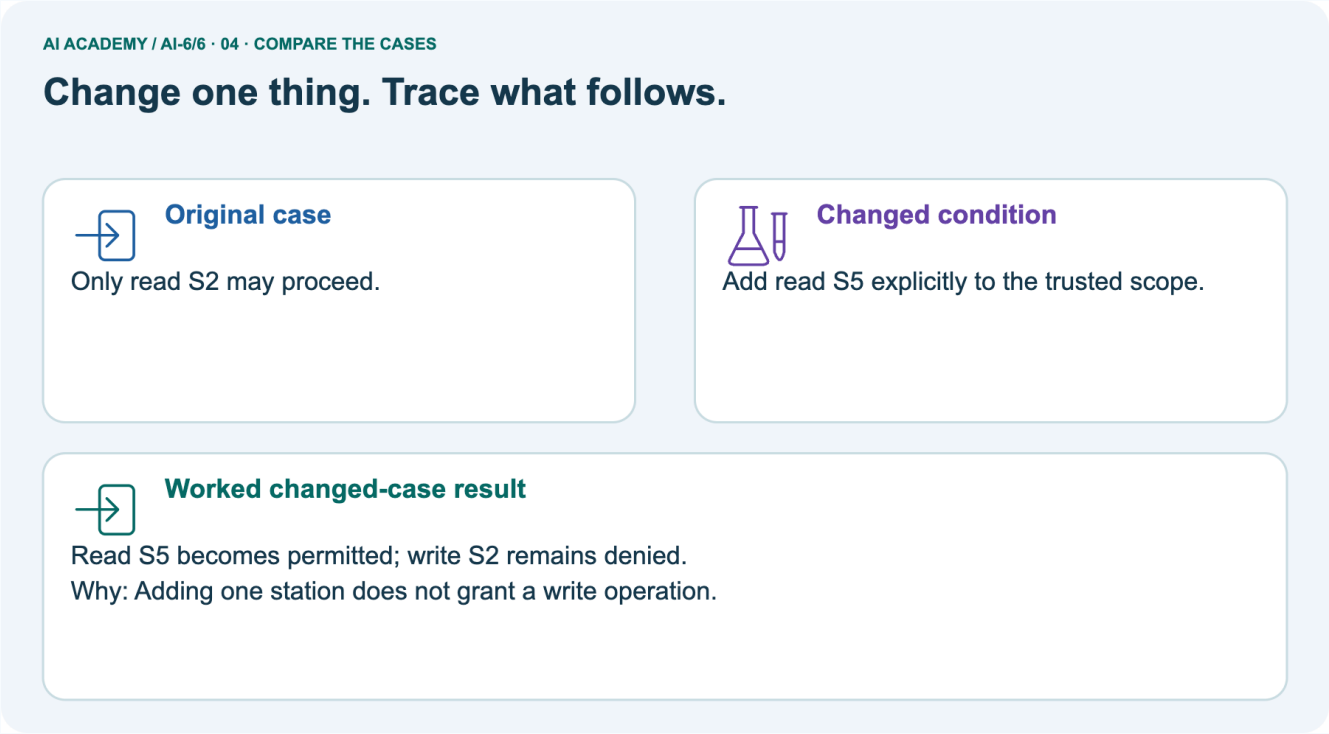
A proposed call writes teacher notes. | A reporting assistant may lookup public rooms but may not write records, send messages or read teacher notes.

Check actor, operation and resource before executing a proposed call. | Both operation and resource exceed scope; deny before execution, regardless of generated confidence.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Only read S2 may proceed. Changed condition: Add read S5 explicitly to the trusted scope. Expected result: Read S5 becomes permitted; write S2 remains denied.

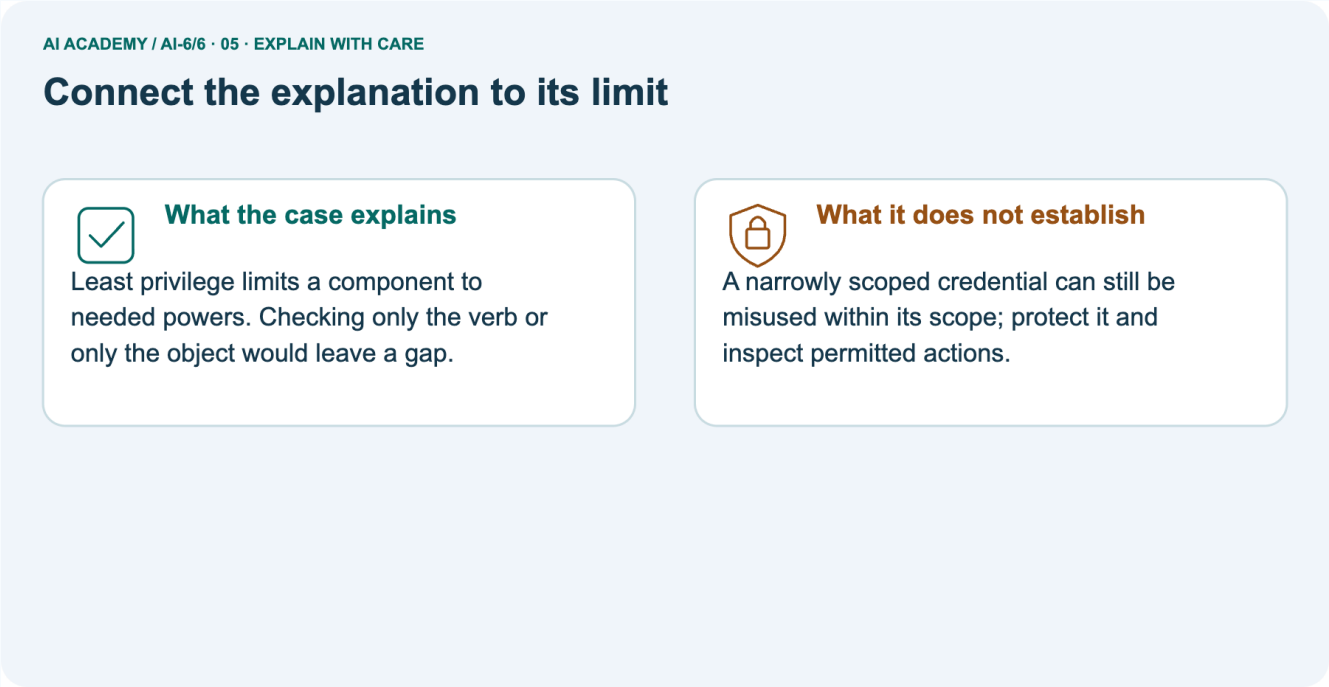
Reason: Adding one station does not grant a write operation.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Adding one station does not grant a write operation.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Least privilege limits a component to needed powers. Checking only the verb or only the object would leave a gap.

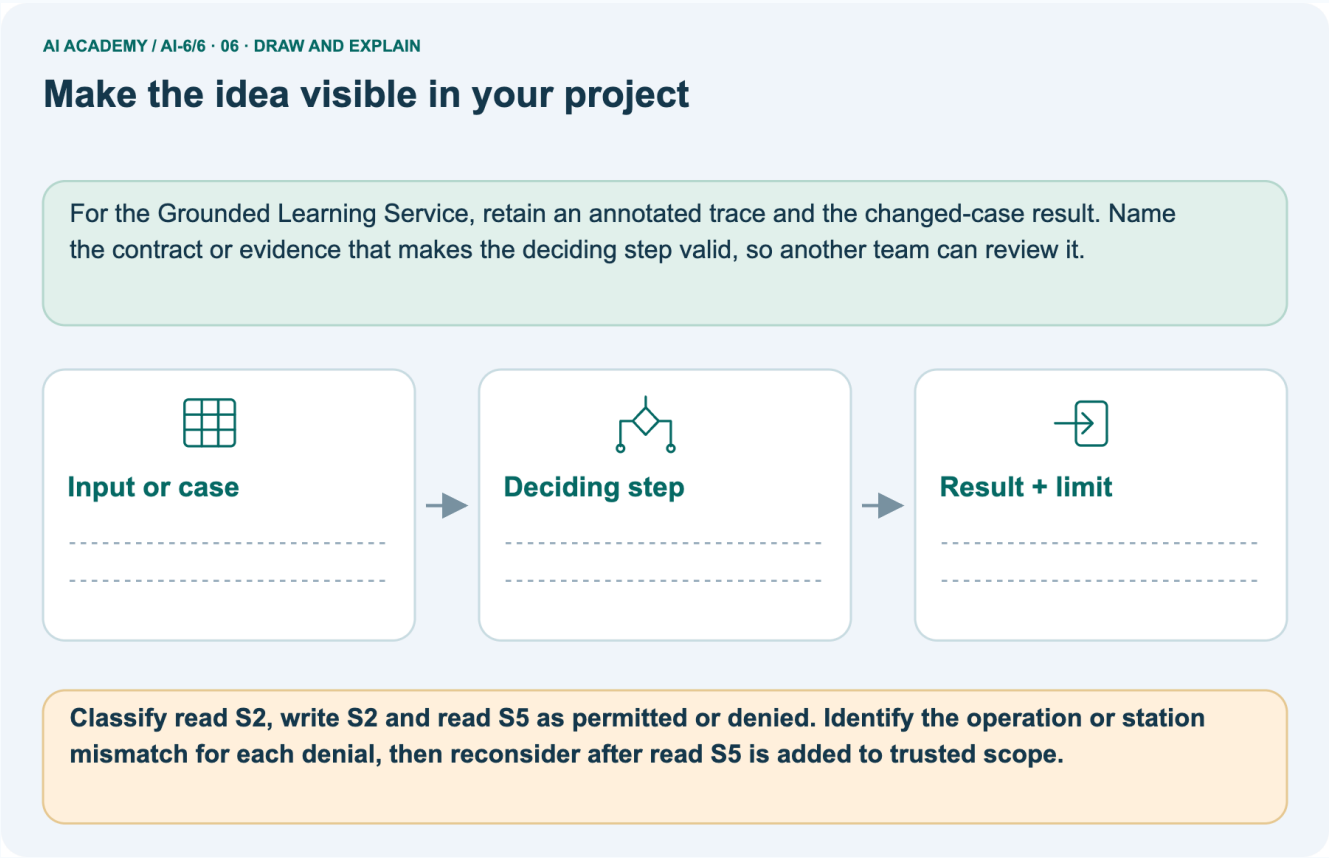
Limit: A narrowly scoped credential can still be misused within its scope; protect it and inspect permitted actions.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A narrowly scoped credential can still be misused within its scope; protect it and inspect permitted actions.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Classify read S2, write S2 and read S5 as permitted or denied.

Identify the operation or station mismatch for each denial, then reconsider after read S5 is added to trusted scope.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Read S2 matches both the allowed operation and station. Write S2 has the wrong operation; read S5 has the wrong station. Only read S2 may proceed.

Changed case: Read S5 becomes permitted; write S2 remains denied. Adding one station does not grant a write operation.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Least privilege and tool scopes: the case

02. Follow the deciding steps

03. Least privilege and tool scopes — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.